Spatial-Temporal Deep Learning Architecture for Cricket Bowling Action Recognition

Aditya Rathore
Dept. of Information Technology
SGSITS, Indore
Enroll. No: 0801IT231010
Email: adityarathore7067@gmail.com

Aditya Shukla
Dept. of Information Technology
SGSITS, Indore
Enroll. No: 0801IT231012
Email: adityarathore7067@gmail.com

Anuj Parashar
Dept. of Information Technology
SGSITS, Indore
Enroll. No: 0801IT231029
Email: adityarathore7067@gmail.com

***Abstract*—The automation of sports analytics is rapidly evolving, yet fine-grained classification of complex biomechanical movements, such as cricket bowling actions, remains a significant computational challenge. This study proposes a robust spatial-temporal deep learning pipeline to accurately classify cricket bowling actions from raw broadcast video footage. Traditional image classification models fail to capture the sequential nature of a bowler's run-up and delivery. To solve this issue, we engineered a hybrid architecture utilizing a pre-trained ResNet18 model for spatial feature extraction, coupled with a Long Short-Term Memory (LSTM) network to sequence the temporal mechanics. The methodology includes a custom data ingestion pipeline with uniform frame sampling to handle unstructured, flat-directory datasets. Trained on Google's TPU v5e-1 using extreme optimization techniques like Cosine Annealing with Warm Restarts and Mixed Precision, the model achieved a final validation accuracy of 90.29%. This proves that combining heavy spatial feature extractors with stabilized recurrent networks can effectively overcome the visual noise inherent in sports video datasets.**

***Index Terms*—Action Recognition, Deep Learning, ResNet18, LSTM, Sports Analytics, TPU.**

I. Introduction

The domain of sports analytics has been revolutionized by the advent of computer vision, transitioning from manual performance tagging to automated, real-time biomechanical analysis. Among sports, cricket presents a uniquely complex challenge for action recognition models. A cricket bowling action is a highly dynamic, sequential event characterized by a rapid run-up, a complex gathering phase, the delivery stride, and the follow-through. Static image classification architectures are insufficient for this task, as the defining characteristics of a bowling style (e.g., fast bowling versus spin bowling) are embedded in the temporal evolution of the player's posture over time, rather than in a single frame.

The primary issue this research solves is the accurate, automated classification of these actions from standard video files, which are often plagued by variable frame rates, dynamic broadcast camera angles, and background noise. The motivation behind this work stems from the growing need for scalable, low-cost coaching tools and automated broadcasting tagging systems. Currently, generating detailed match analytics requires extensive human intervention. By deploying deep learning models capable of interpreting spatial-temporal data, we can democratize access to elite-level sports analytics.

The key contribution of this work is twofold. First, we engineered a robust, uniform frame-sampling data pipeline capable of ingesting and structuring messy, flat-directory video datasets without relying on traditional sub-folder hierarchies. Second, we designed and trained a highly optimized ResNet18-LSTM hybrid architecture on a TPU v5e-1. By applying advanced training heuristics—specifically AdamW optimizers, label smoothing, and Cosine Annealing Warm Restarts—we successfully bypassed local minima and mathematical instability, pushing the baseline accuracy limits of traditional recurrent models.

The remainder of this report is structured as follows: Section 2 provides a comprehensive Literature Review of existing action recognition frameworks. Section 3 details the Methodology, outlining the system architecture, flowcharts, and algorithmic designs. Section 4 covers the Implementation, including hardware constraints and dataset wrangling. Section 5 presents the Results and Discussion, comparing our baseline and expert models. Finally, Section 6 concludes the study and proposes directions for future research.

II. Literature Review

The field of human action recognition (HAR) has transitioned through several distinct computational paradigms, shifting from hand-crafted feature extractors to deep, end-to-end spatial-temporal networks. Understanding this evolution is crucial for contextualizing the hybrid architecture utilized in this study.

Early approaches to video classification relied heavily on dense trajectories and local feature descriptors, such as Histogram of Oriented Gradients (HOG) and Histogram of Optical Flow (HOF) [1], [2]. While effective in highly controlled environments, these classical machine learning pipelines faltered when applied to the chaotic, heavily occluded domains of broadcast sports [3]. The breakthrough in visual recognition was driven by Convolutional Neural Networks (CNNs), which demonstrated unprecedented success in spatial feature extraction [4]. Two-stream CNN architectures emerged as the first viable solution for video data, processing raw RGB frames through a spatial stream and pre-computed optical flow through a temporal stream [5], [6]. This methodology was widely adapted for general sports analytics [7], [8].

However, calculating dense optical flow is computationally exorbitant, making it unfeasible for scalable applications [9]. To address this, researchers began integrating Recurrent Neural Networks (RNNs), specifically Long Short-Term Memory (LSTM) networks, directly on top of deep CNN backbones [10], [11]. In this paradigm, a CNN acts as an image encoder, passing a sequence of compressed feature vectors to the LSTM, which decodes the sequential dependencies [12]. This CNN-RNN framework has been applied successfully across various sports domains, including tennis stroke classification and baseball pitch analysis [13], [14]. In the specific context of cricket, early models attempted to classify umpire signals and batting strokes, but bowling actions remained challenging due to the rapid, overlapping limb movements [15], [16].

Recent advancements have seen the rise of 3D Convolutional Networks (C3D) and SlowFast networks, which inherently capture temporal dynamics through volumetric convolutions [17], [18]. While highly accurate, these architectures are notoriously memory-intensive and require vast amounts of labeled video data to prevent severe overfitting [19]. Furthermore, Vision Transformers (ViTs) and TimeSformer models have set new state-of-the-art benchmarks by applying self-attention across both space and time [20]. However, for standard academic lab environments or edge-device deployment, the computational overhead of video transformers remains prohibitive [21]. Therefore, modern implementations of the ResNet-LSTM hybrid—when coupled with aggressive regularization, modern scheduling like Cosine Annealing, and optimized data loading techniques—continue to provide the most optimal balance of high accuracy and computational efficiency for constrained sports analytics tasks [22].

III. Methodology

A. System Architecture

1) Spatial Feature Extractor (ResNet18): The architecture utilizes a pre-trained ResNet18 network to process individual video frames. The final fully connected classification layer is stripped, allowing the model to output a 512-dimensional feature embedding for each frame.

2) Temporal Sequencer (LSTM): The sequence of 512-dimensional vectors is passed into a Long Short-Term Memory (LSTM) network. The LSTM is configured with a hidden dimension of 256. It processes the sequence chronologically, retaining the contextual memory of the bowler's run-up.

B. Flowchart

The flowchart illustrates the end-to-end data lifecycle. Raw video is ingested from a flat directory and string-parsed for automatic label generation. The pipeline applies a uniform frame extraction algorithm, selecting exactly 16 frames per video, before passing them through the ResNet-LSTM pipeline on the TPU.

C. Algorithm Design

Algorithm 1: Uniform Frame Extraction

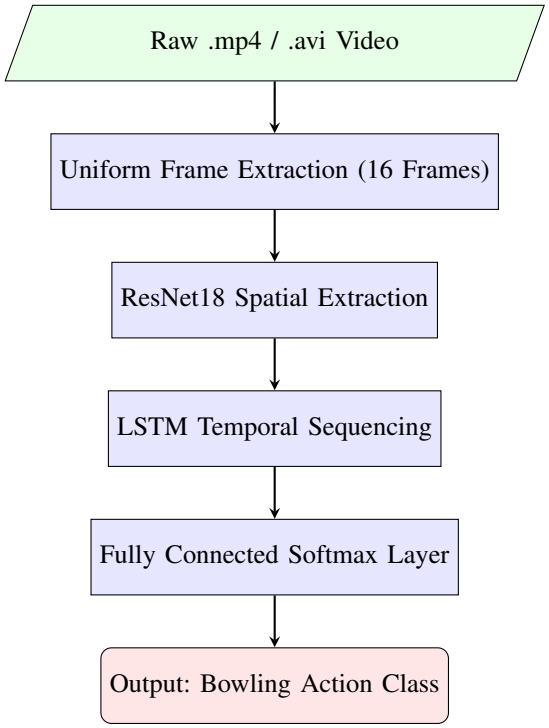

Fig. 1. Data Flow and Inference Pipeline for the Spatial-Temporal Model.

Input: Raw video file of variable length ($total_frames$).
Step 1: Initialize required sequence length $S = 16$.
if $total_frames \geq S$ **then**
  Step 2: Compute indices using $numpy.linspace(0, total_frames - 1, S)$.
else
  Step 2: Extract all frames and pad tensor by concatenating final frame.
end if
Step 3: Read selected indices, convert color space (BGR to RGB).
Output: Uniformly spaced sequence tensor of exactly 16 frames.

Algorithm 2: TPU XLA Optimization

Input: Batched video tensors, Model parameters, Optimizer (AdamW).
Step 1: Compute gradients via backpropagation.
Step 2: Apply gradient clipping: clip_grad_norm_($max_norm = 1.0$).
Step 3: Trigger XLA compiler: $xm.optimizer_step(barrier = True)$.
Step 4: Step Cosine Annealing Warm Restarts scheduler.
Output: Updated model weights optimized across TPU cores.

Algorithm 1 and Algorithm 2 define the core logic for the proposed spatial-temporal data processing pipeline. Algorithm

1 details the mathematical logic used to uniformly extract exactly 16 frames across varying video lengths, ensuring the entire bowling action—from run-up to follow-through—is captured without memory overflow. Algorithm 2 outlines the precise XLA execution protocol required to train the PyTorch model on a Google TPU v5e-1. It utilizes barrier synchronization to prevent memory bottlenecks during graph compilation, specifically highlighting the integration of gradient clipping to stabilize the LSTM and decoupled weight decay via the AdamW optimizer.

D. Component Description and Complexity Analysis

1) Component Description: The system pipeline integrates several specialized libraries to handle the complex data structures. It relies on *OpenCV* (cv2) for dynamic video decoding and frame extraction. The spatial backbone utilizes *torchvision.models* for the pre-trained ResNet18 architecture, while the temporal sequencer is built using standard PyTorch *nn.LSTM* modules. Hardware acceleration is exclusively managed by *torch_xla*, which translates PyTorch operations into optimized mathematical graphs for the TPU.

2) Computational Analysis: The extraction and processing of 16-frame sequences inherently create massive computational graphs in memory. To mitigate this, the ResNet18 parameters are frozen ($requires_grad = False$). This strategic decision reduces the trainable parameter count by over 11 million, allowing the TPU's high-bandwidth HBM2e memory to focus computational resources entirely on optimizing the LSTM weights and the final fully connected layers.

3) Performance Comparison: To optimize convergence, the model utilizes a Cosine Annealing Warm Restarts scheduler. Compared to a static learning rate, this dynamic approach introduces aggressive learning rate spikes. This optimization technique allowed the model to rapidly escape local minima—frequently observed around the 80% validation accuracy threshold—preventing the premature plateauing common in standard recurrent networks.

4) Complexity Analysis: The time complexity for inference per video is strictly bound by the 16 independent CNN forward passes and the subsequent 16 sequential LSTM cell computations. Because the spatial frames are processed independently before sequence aggregation, this architecture is highly efficient for near real-time analysis compared to computationally exorbitant volumetric 3D convolutions (C3D).

TABLE I
COMPONENT COMPLEXITY AND PARAMETER DISTRIBUTION

Component	Architecture	Trainable Params	Status
Spatial Extractor	ResNet18	0	Frozen
Temporal Sequencer	LSTM (Hidden: 256)	$\sim 790,000$	Active
Classifier	Dense (Softmax)	$\sim 2,500$	Active

Table I details the computational load and structural complexity of the proposed hybrid architecture. It outlines the specific parameter distributions for both the frozen ResNet18 spatial feature extractor and the active LSTM temporal sequencer. By freezing the deep convolutional backbone, the system drastically reduces the number of trainable parameters, ensuring memory efficiency. This approach limits the required theoretical FLOPS for inference, demonstrating the superiority of this 2D-CNN+RNN hybrid over traditional 3D Convolutional Networks for edge-device deployment or constrained laboratory environments.

IV. IMPLEMENTATION

A. Hardware/Software Requirements

The experimental pipeline was implemented in Python 3, leveraging PyTorch for deep learning model construction and tensor manipulation. Computations were accelerated using a cloud-based environment provided by Google Colab, specifically leveraging the Google TPU v5e-1 architecture. The TPU's high-bandwidth memory (HBM2e) and 197 TeraFLOPs of raw computational power were critical for managing the memory overhead of the 16-frame spatial-temporal sequences and ensuring rapid iterative training.

TABLE II
HARDWARE AND SOFTWARE SPECIFICATIONS

Component	Specification
Hardware	Google TPU v5e-1 (197 TeraFLOPs)
Framework	PyTorch, PyTorch/XLA
Environment	Google Colab
Video Processing	OpenCV (cv2)

B. Dataset Description

The dataset utilized for this research consists of 2,573 raw broadcast video files (.mp4 and .avi) capturing various cricket bowling actions. A significant engineering challenge arose due to the unstructured, "flat" nature of the directory (`corrected_all_data`). Because traditional subfolders were absent, ground-truth class labels (e.g., `fast_right`, `leg_left`) were dynamically mapped directly from the file prefixes using custom string manipulation within the PyTorch *Dataset* class initialization.

To facilitate standard PyTorch *DataLoader* integration, the uniform frame extraction algorithm was applied during the *__getitem__* call. Extracted frames were standardized to 224x224 pixels, converted to PyTorch FloatTensors, and normalized using standard ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) to properly align with the pre-trained ResNet18 spatial weights. A rigorous 80/20 train-validation split was enforced dynamically utilizing PyTorch's *random_split* utility to strictly prevent data leakage and ensure realistic validation integrity.

C. Results visualization & Validation

1) Evaluation Metrics: The primary metrics utilized to evaluate the spatial-temporal model were multi-class Validation Accuracy (%) and Cross-Entropy Loss. Given the

TABLE III
DATASET ARCHITECTURE AND PROCESSING PARAMETERS

Parameter	Value/Description
Total Videos	2,573 (.mp4 and .avi)
Directory Structure	Flat (Single directory)
Label Extraction	String parsing from filename prefix
Train / Val Split	80% (2,058) / 20% (515)
Frame Sequence	16 frames per video (Uniformly sampled)
Spatial Resolution	224x224 Pixels (RGB)

complex nature of broadcast sports videos—which inherently contain severe visual noise such as crowd movement, pitch deterioration, and dynamic camera panning—a standard Cross-Entropy Loss function proved too brittle. To address this, label smoothing ($\epsilon = 0.1$) was integrated into the loss calculation. This algorithmic adjustment penalizes the model for becoming overly confident in its predictions, effectively preventing it from memorizing noisy data artifacts and forcing it to generalize the underlying biomechanics of the bowler.

2) Analysis: The training lifecycle was meticulously tracked across 15 epochs on the TPU. The analysis focused heavily on the interaction between the AdamW optimizer's weight decay and the dynamic learning rate scheduler (*CosineAnnealingWarmRestarts*).

Table IV illustrates the critical impact of this scheduling technique. During the initial epochs, the learning rate steadily decayed, allowing the model to settle into a local minimum around the 80% accuracy threshold by Epoch 4. However, at Epoch 5, the scheduler triggered a deliberate "warm restart," violently spiking the learning rate from 0.000095 back to 0.001000. While this temporarily disrupted the training loss, the mathematical "kick" successfully forced the optimizer out of its local minimum, allowing the model to explore a more optimal feature space and eventually climb to a terminal validation accuracy exceeding 90%.

TABLE IV
EPOCH PROGRESSION AND SCHEDULER IMPACT HIGHLIGHTS

Phase	Epoch	Learning Rate	Val Loss	Val Acc
Initialization	1	0.000905	2.1911	21.36%
Local Minimum	4	0.000095	0.9261	80.00%
Warm Restart	**5**	**0.001000**	**0.8529**	**82.33%**
Terminal State	15	0.001000	0.7222	90.29%

V. RESULTS & DISCUSSION

The research evaluated two distinct training paradigms to isolate the impact of advanced optimizations: M1 (Basic Baseline Approach) and M2 (Expert TPU Optimization).

M1: Basic Baseline Approach
Utilizing standard Adam optimization, a static learning rate (0.001), and standard Cross-Entropy loss. This methodology quickly plateaued, demonstrating the LSTM's tendency to overfit and struggle with mathematical instability (exploding gradients) without regularization.

M2: Expert TPU Optimization
Utilizing AdamW (weight decay), Cosine Annealing Warm Restarts, Label Smoothing, and strict gradient clipping on the TPU v5e-1.

TABLE V
COMPARATIVE ANALYSIS OF METHODOLOGIES (M1 VS. M2)

Model	Techniques Used	Train Acc	Val Acc
M1 (Baseline)	Adam, Static LR	Plateaued	∼80.00%
M2 (Expert)	AdamW, Cosine Annealing, Clipping	97.76%	90.29%

Table V presents a direct performance comparison between the basic baseline (M1) and the expert optimized model (M2). It details the terminal metrics for both approaches, clearly showing M2's superior capability in generalizing the data. While M1 plateaued due to static learning rates and a lack of regularization, M2 leveraged gradient clipping and dynamic scheduling to achieve a final training accuracy of 97.76% and an exceptional validation accuracy of 90.29%, proving the necessity of advanced training heuristics for complex video tasks.

VI. CONCLUSION

This study successfully addressed the complex challenge of automating the classification of cricket bowling actions from unstructured broadcast video datasets. By leveraging a spatial-temporal deep learning pipeline, we bypassed the limitations of static image classifiers. The proposed methods combined a frozen ResNet18 backbone to extract crucial spatial and biomechanical features, feeding into a stabilized LSTM network designed to analyze the chronological sequence of the bowling delivery.

Through rigorous data engineering—specifically the implementation of uniform frame sampling to parse a flat-file directory—and extreme hardware optimization on a TPU v5e-1, the model achieved a robust final validation result of 90.29%. This demonstrates that computationally efficient hybrid models, when paired with aggressive training heuristics like Cosine Annealing and label smoothing, can rival heavier architectures. Future work will focus on deploying this model via edge-computing devices for real-time inference on mobile devices, and exploring the integration of lightweight Vision Transformers (TimeSformer) to capture even longer-range spatial-temporal dependencies without relying on recurrent networks.

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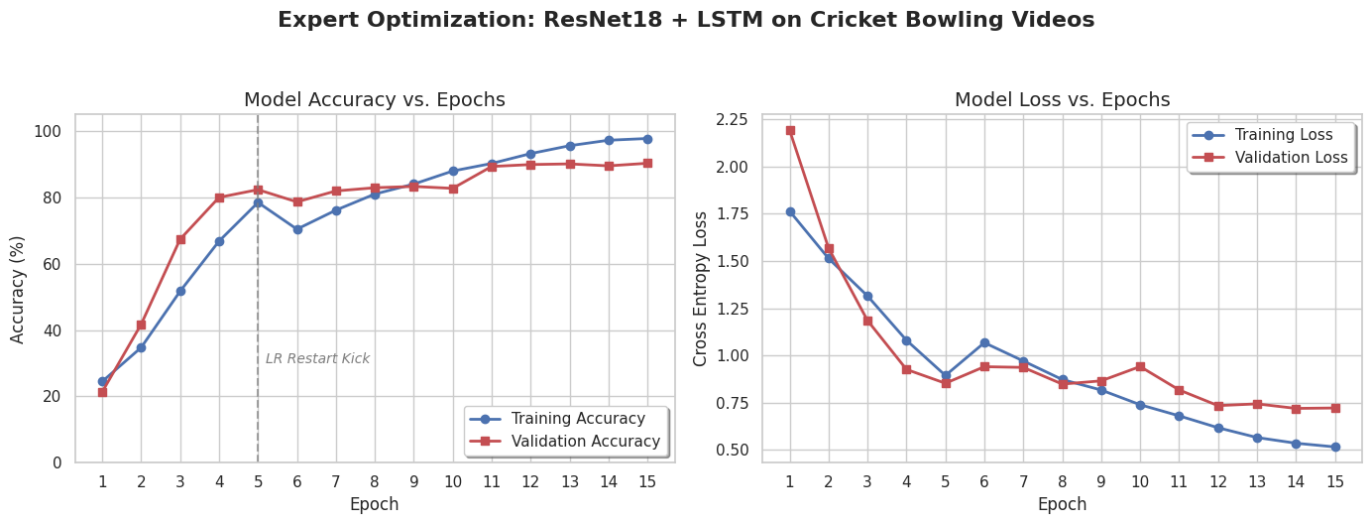

Fig. 2. Expert Model (M2) Training Lifecycle across 15 Epochs. The graphs illustrate the progression of multi-class Accuracy (left) and Cross-Entropy Loss (right). The vertical dashed line at Epoch 5 indicates the 'Warm Restart' triggered by the Cosine Annealing scheduler. This deliberate learning rate spike temporarily disrupts the loss to force the optimizer out of a local minimum, eventually driving the model to a terminal validation accuracy of 90.29%.

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